Dell AI Factory with NVDIA Including NVIDIA GPUs and Spectrum-X Networking Platforms

Abstract

Dell AI Factory with NVIDIA delivers a validated reference architecture combining Dell PowerEdge servers, NVIDIA GPUs, Spectrum-X Ethernet networking platform, NVIDIA AI Enterprise, and Dell PowerScale storage for scalable, high-performance AI infrastructure.

Dell AI Platform Design Guide

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Notes, cautions, and warnings

NOTE: A NOTE indicates important information that helps you make better use of your product.

CAUTION: A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

WARNING: A WARNING indicates a potential for property damage, personal injury, or death.

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Introduction

Topics:

- Executive summary
- Overview
- Document purpose
- Audience
- Revision history

Executive summary

The Dell AI Factory with NVIDIA delivers a validated architecture that integrates scalable, accelerator-optimized compute infrastructure with a low-latency, high-performance networking platform to meet the demands of modern generative AI workloads. By combining Dell PowerEdge servers with NVIDIA GPUs, the NVIDIA Spectrum-X Ethernet networking platform, NVIDIA AI Enterprise software, and Dell PowerScale storage, the solution enables enterprises to efficiently train, fine-tune, and deploy LLMs using resilient, predictable, cost-effective, on-premises infrastructure. It is also designed to be easily deployed and managed, helping organizations accelerate time to value while reducing complexity.

Overview

Generative AI introduces demanding requirements for performance, scalability, and reliability as model sizes and datasets continue to grow. Achieving efficient training, fine-tuning, and inference across distributed systems requires high-performance GPU-accelerated compute and low-latency networking. This design guide outlines the architectural principles and reference design that enable enterprises to support these AI workloads with consistent performance at scale.

The solution incorporates a comprehensive AI software platform that streamlines provisioning, lifecycle management, and orchestration of GPU-accelerated environments. It provides the runtime components and tooling required for distributed LLM training, fine-tuning, and model deployment. By automating complex infrastructure tasks and standardizing the software environment, it helps ensure consistent performance, efficient resource use, and simplified operations for managing generative AI workloads at enterprise scale.

This design also aligns with NVIDIA's Enterprise Reference Architecture, which defines validated system configurations and best practices for building scalable, predictable, multi-node GPU environments. By applying these reference patterns across PowerEdge servers, the solution follows proven guidelines for compute, networking, and bandwidth, ensuring predictable performance and reliable operation. The remainder of this document details how the architecture, software stack, and networking design follow these principles.

Document purpose

This document provides validated architectural guidance and configuration recommendations for deploying enterprise-grade generative AI using Dell infrastructure with NVIDIA technologies.

Audience

This design guide is intended for technical professionals responsible for planning, architecting, deploying, and operating enterprise generative AI solutions. It is written for AI architects, IT infrastructure architects, system administrators, platform engineers, and operations teams who require guidance on building scalable, high-performance environments for LLM training, fine-tuning, and inference. The content also supports data scientists and AI engineers who need to understand the underlying infrastructure required to run advanced GPU-accelerated workloads effectively. A working knowledge of AI workflows, large-scale compute environments, and core generative AI concepts is assumed.

Revision history

The following table lists the revision history.

Table 1. Revision history

Date	Version	Change summary
November 2024	H20214.0	Initial release.
January 2025	H20214.1	Addition of the NVIDIA H200 SXM GPU on the Dell PowerEdge XE9680 and the NVIDIA H100 NVL GPU on the Dell PowerEdge R760xa server. Update to NVAIE 5.2 software.
May 2025	H20214.2	Support for NVIDIA AI Enterprise 6.2 and Ubuntu 24.04. Support for NVIDIA Enterprise Reference Architecture for PowerEdge R76xa-based cluster.
July 2025	H20214.3	Addition of Run:ai for GPU orchestration.
October 2025	H20214.4	Addition of Dell AI optimized 17G servers – PowerEdge XE7740 and XE7745 servers with support for NVIDIA RTX Pro 6000 Blackwell Server Edition and NVIDIA H200 HVL. Updated network and software architecture.
December 2025	H20214.5	Updated Table 2 PERC to H965e. Updated the memory configuration in Table 3.
January 2026	H20214.6	Added support for PowerEdge Datacenter servers for Inference only use cases. Updated NVIDIA Spectrum SN5600 switches to the newly released Spectrum SN5610.
April 2026	H20214.7	Revised the document structure to better align with the supported components. Removed PowerEdge R760xa configuration, since it is now retired.

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Solution architecture

The solution architecture defines the core building blocks and design principles of the Dell AI Factory with NVIDIA, outlining the compute, networking, storage, and software components that work together to deliver a scalable, high-performance platform for enterprise generative AI workloads.

Topics:

- Design principles
- Reference architecture
- Solution architecture components
- NVIDIA AI Enterprise
- Security considerations

Design principles

Building an effective generative AI platform requires a clear set of guiding principles that ensure the architecture delivers high performance, operational efficiency, and flexibility across different deployment sizes and workload types. The following principles are used in the design of the Dell AI Factory with NVIDIA.

- **Alignment with NVIDIA Enterprise Reference Architecture (ERA)**

  The solution incorporates validated design patterns and configuration guidance from NVIDIA ERA to take advantage of proven methods for multi-node GPU deployments and optimal system configuration.

- **High-performance GPU backend (optional for inference-only)**

  Training and fine-tuning workloads benefit from a dedicated GPU interconnect built on the NVIDIA Spectrum-X Ethernet platform and SuperNICs, enabling fast east-west traffic and efficient multi-node communication. Inference-only environments can forgo this backend network.

- **Modular and customizable architecture**

  The reference design is intentionally modular, allowing organizations to adapt components such as the management stack, storage platform, or cluster size and composition. Customers can integrate Dell PowerScale storage, Dell AI Data Platform and extend the architecture as their needs evolve.

- **Container-native, service-oriented platform**

  Kubernetes provides a unified orchestration layer that simplifies deployment, maintenance, and resource management. Its service-oriented design supports flexible hosting of training, fine-tuning, and inference workflows while enabling operational automation.

- **Enterprise-class availability and resilience**

  The networking and system layout incorporate redundancy, robust failover mechanisms, and traffic separation to maintain reliability across management, storage, and GPU fabrics.

- **Support for varied workload sizes and cluster footprints**

  The architecture accommodates small inference clusters as well as larger multi-node GPU environments, making it suitable for a wide range of generative AI use cases.

- **Streamlined operations and lifecycle management**

  Through standardized components and automated orchestration, the platform reduces administrative overhead and helps maintain uniform operational practices across the AI development and deployment lifecycle.

Reference architecture

The reference architecture integrates compute, networking, storage, orchestration, and AI software capabilities into a unified platform for enterprise generative AI. It supports a wide range of use cases—including RAG, AI agents, code assistants, fine-tuning, and inference—while giving organizations the flexibility to scale their environment and adapt components to their operational needs. The following figure shows the key components of the reference architecture:

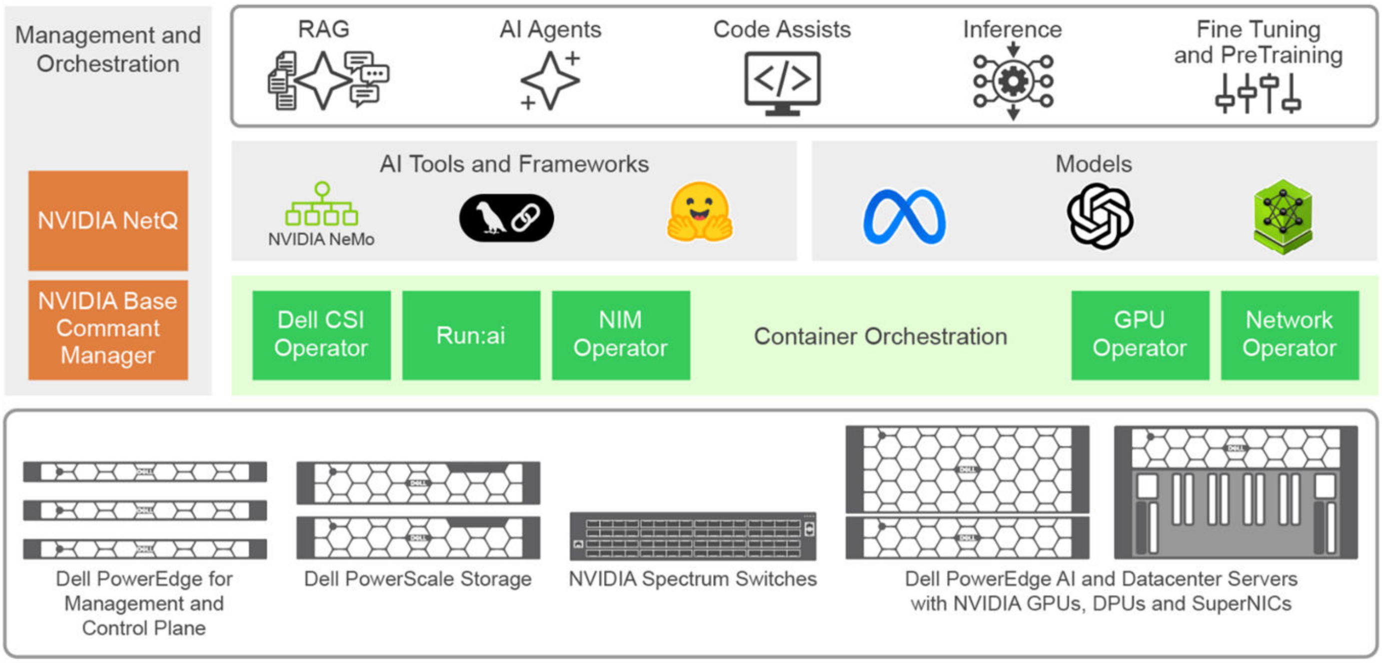

Figure 1. Reference architecture Dell AI Factory with NVIDIA using Spectrum-X Ethernet Networking Platform

Compute and management hardware

Dell provides a range of server platforms and GPU options that can be selected based on workload requirements. High-density, AI-optimized servers are designed for distributed training, fine-tuning, and other compute-intensive tasks where a dedicated GPU backend network is beneficial. Datacenter-class servers deliver a cost-efficient option for inference-only deployments that do not require backend GPU connectivity. Management and control-plane services run on separate general-purpose servers to ensure operational stability and clean separation of responsibilities.

Networking

The architecture uses the NVIDIA Spectrum-X networking platform, built on Spectrum-4 switches and NVIDIA SuperNICs, to deliver a single, consolidated fabric that supports both frontend and backend traffic. Frontend services—including management, storage, and application connectivity—use EVPN-VXLAN to provide enterprise-class segmentation, multihoming, and flexible L2/L3 connectivity across the fabric. The backend network enables high-bandwidth, low-latency east-west GPU communication, supporting efficient distributed training and fine-tuning across multi-node environments. All switches run NVIDIA Cumulus, a Linux-based network operating system that provides automation, telemetry, and operational consistency.

Management and orchestration

NVIDIA Base Command Manager Essentials automates provisioning, cluster bring-up, and lifecycle workflows, simplifying deployment and day-to-day operations. NVIDIA NetQ adds real-time network visibility, validation, and troubleshooting capabilities, helping ensure the fabric operates reliably and according to recommended design patterns.

Container orchestration and operators

Kubernetes provides the orchestration layer for deploying and managing AI workloads across the cluster. The platform integrates several operators, including GPU Operator for driver and runtime management, Network Operator for RDMA and GPUDirect enablement, Dell CSI Operator for storage provisioning, and Run:ai for advanced workload scheduling and GPU resource allocation. These components work together to deliver a fully cloud-native, automated environment for training, fine-tuning, and inference.

Storage

This solution uses Dell PowerScale storage as a repository for datasets for model customization, models, model versioning and management, and model ensembles.

The flexible and robust storage capabilities of PowerScale offer the scale and speed necessary for training and operationalizing AI models, providing a foundational component for AI workflow. Its capacity to handle the vast data requirements of AI, combined with its reliability and high performance, cements the crucial role that external storage plays in successfully bringing AI models from conception to application.

There are several models used in generative AI solution architectures, such as PowerScale F710, all powered by the PowerScale OneFS operating system and supporting inline data compression and deduplication. The minimum number of PowerScale nodes per cluster is three nodes. The maximum cluster size is 252 nodes, which supports up to 186 PB of raw capacity.

AI frameworks and models

The solution supports a broad set of AI frameworks and model toolchains used for LLM training, fine-tuning, and inference. These frameworks enable organizations to build, adapt, and deploy models efficiently using optimized, enterprise-ready software components.

Use cases

The architecture is designed to support key generative AI scenarios such as retrieval-augmented generation (RAG), agent-based applications, code assistants, model fine-tuning, and scalable inference. These workloads benefit from the platform's compute density, networking performance, and integrated software ecosystem.

Solution architecture components

To provide a concise view of the Dell AI Factory design, the following table summarizes the key components that make up the reference architecture. This includes the management plane, compute options, networking fabric, storage subsystem, orchestration layer, and the software stack. Worker-node options are presented together to show how customers can tailor the cluster based on training, fine-tuning, or inference-only requirements.

Table 2. Reference architecture summary

Component	Configuration
Management Server	4 × PowerEdge R670
Compute Nodes	Options include: ● Up to 16 × PowerEdge XE7740or XE7745 ● Up to 12 x PowerEdge XE9680 ● Up to 32 x PowerEdge R770 / R7725 / R7715 (inference-only)
Networking	● 2 × Spectrum-4 SN5610 (converged frontend + backend) ● 1 × Spectrum SN2201 (OOB management)
Storage	PowerScale F710
Cluster Manager	Base Command Manager
Container Orchestration	Upstream Kubernetes
AI Frameworks & Software	NVIDIA AI Enterprise

NVIDIA Enterprise Reference Architecture

NVIDIA's Enterprise Reference Architectures (ERAs) provide standardized, validated design guidance for building reliable, high-performance AI infrastructure. They offer clear recommendations for system configuration, network design, and component selection, helping organizations reduce deployment risk and accelerate time to value. By following ERA guidance, partners and customers benefit from optimized performance, improved utilization, higher uptime, and predictable operational characteristics across AI Factory deployments.

NVIDIA uses a structured methodology to introduce ERAs for each new technology generation—including Hopper and Blackwell GPUs, Grace CPUs, Spectrum-X networking, and BlueField-based platforms. Each ERA includes configuration guides and best practices to help partners design and deliver optimized systems for enterprise-class AI workloads. All ERAs are anchored on NVIDIA-Certified systems and follow a standardized "Reference Configuration" format based on four parameters:

C: Number of CPU sockets

G: Number of GPUs

N: Number of network adapters (NICs)

B: Average East-West network bandwidth per GPU (in GbE)

This C-G-N-B model simplifies system selection and ensures balanced performance across compute, networking, and I/O.

Table 3. NVIDIA ERA configurations and its mapping to Dell PowerEdge servers

ERA Configuration (C-G-N-B)	Related Dell servers
2-4-3-200	PowerEdge R760xa (retired. Not in scope of this document)
2-8-9-400	PowerEdge XE9680
2-8-5-200	PowerEdge XE7740 and PowerEdge XE7745
2-8-9-800 (Single and Dual Plane)	PowerEdge XE9780 and PowerEdge XE9785 (Upcoming)

NVIDIA review and endorsement—NVIDIA performs a vendor-level review of partner designs to ensure alignment with ERA guidelines. When a partner's architecture meets the required configuration, bandwidth, and system-level criteria, NVIDIA formally endorses the design as ERA-aligned. The Dell configurations corresponding to ERA 2-8-9-400 and 2-8-5-200 have been reviewed and endorsed by NVIDIA. For more information, refer to NVIDIA's Enterprise Reference Architecture Partner Endorsed Designs.

Inference-only designs using Dell datacenter servers (without GPU backend networking) are **not** part of NVIDIA's ERA program and therefore are not ERA-endorsed.

NVIDIA AI Enterprise

NVIDIA AI Enterprise provides the full-stack software foundation for the Dell AI Factory, delivering an integrated suite of tools that streamline the development, customization, deployment, and operation of AI workloads. It supplies the enterprise-grade software components needed to run training, fine-tuning, inference, and data-driven AI services across a standardized and operationally consistent platform.

At the infrastructure layer, Base Command Manager Essentials automates provisioning, node initialization, and lifecycle workflows for the Dell AI Factory, ensuring repeatable deployments and reducing the operational overhead of standing up and maintaining GPU-accelerated clusters.

NVIDIA AI Enterprise also delivers Kubernetes-native operators that automate and optimize the underlying infrastructure within the Dell AI Factory. The GPU Operator manages driver installation, CUDA runtimes, monitoring components, and GPU readiness across all nodes. The Network Operator enables advanced data-path features such as RDMA and GPUDirect, ensuring optimal performance for multi-node distributed training. The NIM Operator simplifies deployment of accelerated inference microservices, allowing organizations to serve optimized foundation models with minimal operational effort.

For workload scheduling, resource allocation, and multi-tenant sharing, Run:ai is included as part of the software stack. Within the Dell AI Factory, Run:ai enhances GPU utilization by enabling fractional GPU allocation, prioritization, and intelligent workload placement across teams and projects.

NVIDIA NeMo Framework is a scalable and cloud-native generative AI framework that is built for researchers and developers working on Large Language Models, Multimodal, and Speech AI (for example, Automatic Speech Recognition and Text-to-Speech). It enables users to efficiently create, customize, and deploy new generative AI models by leveraging existing code and pretrained model checkpoints.

Together, these components form the software backbone of the Dell AI Factory—providing a cloud-native, automated, and secure operational model that accelerates enterprise adoption of generative AI while simplifying ongoing infrastructure management.

Security considerations

AI platforms exist at the intersection of rapidly evolving AI technology and the ever-challenging security landscape. This unique position brings both incredible opportunities and significant risks. The security of any system exists in relation to the system's threat profile: the value of a system's assets, the likelihood of an attempted attack, and the nature of potential attackers in terms of their resources, capabilities, and incentives. Furthermore, there is often a tension between security and convenience. Extra security measures might interfere with desirable use cases or lead to false-positive results. Therefore, Dell Technologies advises customers and architects to consider their specific operational contexts and conduct appropriate analysis to validate the applicability of the recommendations in this guide. Depending on the threat profile, additional services and security measures might be advisable.

The Dell AI Platforms security best practices guide describes some of the potential threats to an AI Platform and offers guidance on how to mitigate them. It also describes some common security configuration procedures, followed by guidance related to categories of potential threats and their respective mitigations.

The recommendations in the security best practices guide are not a complete security evaluation of the Dell AI Platform in its entirety or its individual components, nor a complete threat and risk analysis. You can find additional information at the Dell Security and Trust Center.

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Management and compute infrastructure

The Dell AI Factory is built on two primary compute infrastructure layers: a management plane that delivers orchestration, control, and operational services, and a set of compute nodes that provide the GPU acceleration required for training, fine-tuning, and inference. The management hardware ensures stable cluster operations, while the compute nodes deliver the performance and scalability required for generative AI workloads.

Topics:

- Management server configuration
- GPU worker node configuration

Management server configuration

The management plane is anchored by four dedicated PowerEdge servers. Three of the servers form the Kubernetes control plane. The fourth server is intended to run VMware ESXi and host virtual machines that provide the operational tooling for the Dell AI Factory. These management virtual machines typically include NVIDIA Base Command Manager Essentials, NVIDIA NetQ, and other supporting components such as observability, logging, or monitoring services.

If a customer already operates a virtual infrastructure, such as an existing VMware or similar enterprise virtualization platform, the fourth server can be used differently. It may serve as a dedicated Base Command Manager node, a login or utility node for administrators, or a host for other management components depending on operational needs.

The following table provides the recommended minimum configuration for the PowerEdge R670 head node and the control plane nodes:

Table 4. Management Server configuration

Component	Configuration	Guidance
Server model	4 x PowerEdge R670	Standardized platform for management services and control plane.
CPU	Intel Xeon 6 Performance 6515P (16 cores, 32 threads)	Sufficient for control-plane and management workloads. Customers may upgrade to higher-core or dual-socket configurations if running a larger virtualization footprint or additional management tooling.
Memory	8 x 32 GB (256 GB) RDIMM, 6400MT/s, Dual Rank	Minimum recommended. Customers can increase memory for expanded monitoring, logging, or VM density.
Boot / OS Storage	BOSS-N1 module with 2 × 960 GB M.2 (RAID 1)	Recommended for OS resilience and simplified recovery.
RAID controller	PERC H965e	Provides resiliency for local storage. Optional if customers rely on external storage arrays for VM datastores.
Local storage	4 × 3.84 TB SAS SSDs (Read-intensive)	Provides capacity for container images, VM disks, and logs. Optional if external storage arrays are used.
Frontend network	NVIDIA Mellanox ConnectX-6 Dx Dual-Port 100 GbE	Provides sufficient bandwidth for in-band management and service traffic.

GPU worker node configuration

Dell Technologies provides a selection of GPU-optimized servers suitable for configuration as worker nodes for generative AI: the PowerEdge XE7740, XE7745, and PowerEdge XE9680 servers. You have the flexibility to choose one of these PowerEdge servers that is based on the specific model size that you require. Larger models, characterized by a greater parameter size, require servers that are equipped with a higher GPU count and enhanced connectivity.

The GPU-optimized servers act as worker nodes in a Kubernetes cluster. The number of servers depends on the size of the model, the customization method, and the end-user requirements for training time. Larger models, characterized by a greater parameter size, require servers that are equipped with a higher GPU count and enhanced connectivity.

PowerEdge XE7740 / XE7745 GPU worker node

The PowerEdge XE7740 and XE7745 server is a two-socket, 4U server that is designed specifically for AI tasks. It supports eight PCIe based accelerators, ideal for machine learning and deep learning training and inferencing workloads, especially for those workloads that are training LLMs.

The following table provides a recommended configuration for PowerEdge XE7740/XE7745 worker node:

Table 5. PowerEdge XE7740 / XE7745 GPU worker node

Component	Details	Guidance
Server model	PowerEdge XE7740 / XE7745	AI-optimized platforms for multi-GPU workloads; customers may choose Intel- or AMD-based variants based on preference.
CPU	Dual-socket CPUs (Intel Xeon 6 on XE7740, AMD EPYC on XE7745), 64 cores per socket	Suitable for large GPU clusters. Higher-frequency or higher-bin CPUs may be used for preprocessing-heavy workflows, but customers must ensure power budgets support those configurations.
Memory	For PowerEdge XE7740 – 32 x 64 GB RDIMM For PowerEdge XE7745 – 24 x 64 GB RDIMM	General guideline: minimum of 1 TB for 8-GPU systems. Fully populated DIMMs improve bandwidth; customers may scale memory up or down based on pipeline needs.
Boot / OS Storage	BOSS-N1 with 2 × 960 GB M.2 (RAID 1)	Standard resilient boot option.
Local storage	2 × 3.84 TB NVMe SSD (Read-intensive)	Provides fast access for local datasets, container images, temp storage, and model artifacts.
GPU (accelerator)	8 × NVIDIA H200 NVL or 8 × NVIDIA RTX Pro 6000 BSE	Choose based on target workload characteristics (training, fine-tuning, or inference).
Frontend network	1 × NVIDIA BlueField-3 Dual-Port 200 GbE (B3220)	This interface is a DPU-based adapter, enabling advanced capabilities such as security offload, firewalling, and accelerated networking services when required. If DPU-based functionality is not needed, customers may instead choose NVIDIA ConnectX-7 adapters as a simpler, NIC-only option while maintaining high-performance network connectivity.
Backend network	4 × NVIDIA BlueField-3 Single-Port 400 GbE (B3140H)	Required for distributed training; **not required** for inference-only deployments.

PowerEdge XE9680 GPU worker node

The PowerEdge XE9680 server is a two-socket, 6U server that is designed specifically for AI tasks. It supports eight accelerators, ideal for machine learning and deep learning training and inferencing workloads, especially for those workloads that are training LLMs.

The PowerEdge XE9680 server with the NVIDIA H200 accelerator offers high-performance capabilities for enterprises seeking to unlock the value of their data and differentiate their business with customized LLMs. The following table provides a recommended configuration for a PowerEdge XE9680 GPU worker node:

Table 6. PowerEdge XE9680 GPU worker node

Component	Details	Guidance and Options
Server model	PowerEdge XE9680	AI-optimized platforms for multi-GPU workloads.
CPU	2 x Intel Xeon Platinum 8562Y+ 2.8G, 32C/64T, 20 GT/s, 60M Cache	Suitable for large GPU clusters. Higher-frequency or higher-bin CPUs may be used for preprocessing-heavy workflows, but customers must ensure power budgets support those configurations.
Memory	16 x 128 GB RDIMM, 5600 MT/s Dual Rank	General guideline: minimum of 1 TB for 8-GPU systems. Customers may scale memory up or down based on pipeline needs.
Operating system storage	BOSS-N1 with 2 × 960 GB M.2 (RAID 1)	Standard resilient boot option.
Local storage	4 × 3.84 TB NVMe SSD (Read-intensive)	Provides fast access for local datasets, container images, temp storage, and model artifacts.
Frontend network	1 × NVIDIA BlueField-3 Dual-Port 200 GbE (B3220)	This interface is a DPU-based adapter, enabling advanced capabilities such as security offload, firewalling, and accelerated networking services when required. If DPU-based functionality is not needed, customers may instead choose NVIDIA ConnectX-7 adapters as a simpler, NIC-only option while maintaining high-performance network connectivity.
GPU (accelerator)	8 x NVIDIA H200 SXM	Standard GPU option with the server.
Backend network	8 × NVIDIA BlueField-3 Single-Port 400 GbE (B3140H)	Required for distributed training; **not required** for inference-only deployments.

PowerEdge Datacenter servers (for inference only)

Dell PowerEdge R770, R7725, and R7715 servers offer a flexible and scalable platform for AI inference workloads. These servers combine high-performance CPUs with support for GPU acceleration, enabling efficient deployment of machine learning models in production environments. The R770, powered by Intel Xeon 6 CPU, delivers balanced compute and storage for enterprise AI, while the R7725, powered by AMD EPYC processors, provides exceptional memory bandwidth and multi-GPU support for demanding inference tasks. Together, these servers are ideal for organizations looking to integrate AI inference into existing data center infrastructure without the need for specialized hardware.

Table 7. PowerEdge Datacenter server GPU worker nodes

Component	PowerEdge R770	PowerEdge R7725	PowerEdge R7715
CPU	2 x Intel® Xeon® 6 Performance 6745P 3.1G, 32C/64T, 24GT/s, 336M Cache, Turbo, (300W) DDR5-6400	2 x AMD EPYC 9335 3.0GHz, 32C/64T, 128 Cache (210W) DDR5-6400	AMD EPYC 9335 3.0GHz, 32C/64T, 128 Cache (210W) DDR5-6400
Memory	16 x 32GB RDIMM, 6400MT/s, Dual Rank	16 x 32GB RDIMM, 6400MT/s, Dual Rank	8 x 32GB RDIMM, 6400MT/s, Dual Rank
BOSS Card	BOSS-N1 controller card + with 2 M.2 960 GB (RAID 1)	BOSS-N1 controller card + with 2 M.2 960 GB (RAID 1)	BOSS-N1 controller card + with 2 M.2 960 GB (RAID 1)
Storage	4 x 480GB SSD SATA Read	4 x 480GB SSD SATA Read	4 x 480GB SSD SATA Read

Table 7. PowerEdge Datacenter server GPU worker nodes (continued)

Component	PowerEdge R770	PowerEdge R7725	PowerEdge R7715
	Intensive 6Gbps 512e 2.5in Hot-plug AG Drive, 1 DWPD	Intensive 6Gbps 512e 2.5in Hot-plug AG Drive, 1 DWPD	Intensive 6Gbps 512e 2.5in Hot-plug AG Drive, 1 DWPD
Frontend network	1 x NVIDIA ConnectX-6 Dx Dual Port 100GbE	1 x NVIDIA ConnectX-6 Dx Dual Port 100GbE	1 x NVIDIA ConnectX-6 Dx Dual Port 100GbE
GPU (accelerator)	2 x NVIDIA H200 NVL GPUs, 2 x NVIDIA RTX Pro 6000 Blackwell Server Edition GPUs, or 2 x NVIDIA L40S GPUs	2 x NVIDIA H200 NVL GPUs, 2 x NVIDIA RTX Pro 6000 Blackwell Server Edition GPUs, or 2 x NVIDIA L40S GPUs	2 x NVIDIA H200 NVL GPUs or 2 x NVIDIA L40S GPUs
Backend network	None	None	None

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Networking design

A high-performance network fabric is one of the most critical components of any AI infrastructure, and it becomes especially important for large-scale generative AI workloads. Distributed training, fine-tuning, and multi-node inference depend on predictable, low-latency communication between GPUs and between compute, storage, and management services. The Dell AI Factory builds on the NVIDIA Spectrum-X Ethernet Networking Platform, which is designed specifically to meet the demands of modern GPU clusters by delivering high throughput, deterministic performance, congestion-free operation, and large-scale efficiency—all while retaining the operational simplicity and ecosystem benefits of Ethernet.

The network architecture adheres to NVIDIA's Enterprise Reference Architecture (ERA) guidelines, ensuring that the fabric aligns with validated design patterns for multi-node GPU deployments. Spectrum-X combines a rail-optimized topology with RoCEv2 acceleration, adaptive routing, and congestion control, enabling AI workloads to scale from a handful of nodes to multi-rack environments without compromising performance. Together, these elements form the backbone of a converged, enterprise-class Ethernet design suitable for data-center-scale AI factories.

Topics:

- Traffic types
- Physical network architecture
- Logical network architecture
- Cables and optics

Traffic types

The Dell AI Factory network supports a diverse set of traffic flows, each with different performance, availability, and isolation requirements. The Spectrum-X converged Ethernet fabric enables these traffic types to coexist efficiently across a shared underlay.

- **GPU compute fabric (east-west)**—This fabric carries distributed-training and collective-communication traffic between GPUs across nodes. This GPU traffic uses RoCEv2 with PFC, adaptive routing, and congestion control to ensure low-latency, lossless east-west communication essential for LLM training and large-scale fine-tuning.
- **Front-end network traffic (north-south)**—Front end traffic includes storage access, application flows, and in band cluster management. Delivered over an EVPN VXLAN overlay, this traffic benefits from workload segmentation, multihoming, and scalable L2/L3 virtualization.
- **Management traffic**
  - **Out-of-band (OOB)** traffic supports iDRAC access, power management, and system-level administration.
  - **In-band** management supports Kubernetes API communication, monitoring, lifecycle operations, and operational tooling.
- **Additional traffic classes**
  - Storage traffic—Communication with PowerScale for datasets, checkpoints, and AI artifacts
  - Edge/north-south traffic—Connectivity to upstream routers, service layers, or enterprise networks
  - EVPN-VXLAN overlay traffic—Virtualized L2/L3 service delivery across the fabric
  - Underlay transport—BGP unnumbered and BFD that provide resilient routing and fast convergence

Physical network architecture

The physical network architecture of the Dell AI Factory is built around a converged Spectrum-X Ethernet fabric that carries both east-west GPU traffic and north-south storage, application, and management traffic across the same pair of Spectrum-4 SN5610 switches. This approach simplifies deployment, reduces cabling complexity, and aligns with NVIDIA ERA design patterns that consolidate network tiers while still providing deterministic performance for AI workloads.

At the GPU layer, compute nodes connect to the fabric through 400 GbE BlueField-3 SuperNICs, which provide the high-bandwidth, low-latency east-west communication necessary for multi-node training and collective operations. These SuperNICs enable deterministic GPU communication by leveraging RDMA/RoCEv2, Priority Flow Control (PFC), and congestion-aware adaptive routing.

For storage, application, and general north-south data flows, each compute and management node connects to the same switches using dual 200 GbE BlueField-3 DPUs. These ports provide resilient, high-throughput access to PowerScale storage and deliver sufficient bandwidth for model checkpoints, dataset movement, and in-band management tasks.

The three Kubernetes control-plane nodes and the management server use 100 GbE uplinks to join the converged fabric, allowing them to communicate with the compute cluster. Similarly, PowerScale nodes attach directly to the SN5610 switches using 200 GbE links, ensuring access to AI datasets.

Out-of-band (OOB) management is handled by a dedicated SN2201 1 GbE switch, which connects all server iDRAC interfaces, BlueField DPU management ports, and switch OOB interfaces.

Network architecture for AI optimized servers with backend networking

The following diagram presents a high-level view of a datacenter network architecture integrating Dell and NVIDIA components. It features PowerEdge R670 servers for Base Command Manager and Kubernetes Control Plane, PowerEdge XE7740/XE7745 servers, PowerScale Storage, and Spectrum-X SN2201 for out-of-band management. Two SN5610 switches serve as the core fabric, interconnecting these elements. The network uses multiple Ethernet types, color-coded for clarity: 400 GbE GPU Ethernet (green), 200 GbE storage and access (orange), 1 GbE Out-of-Band management.

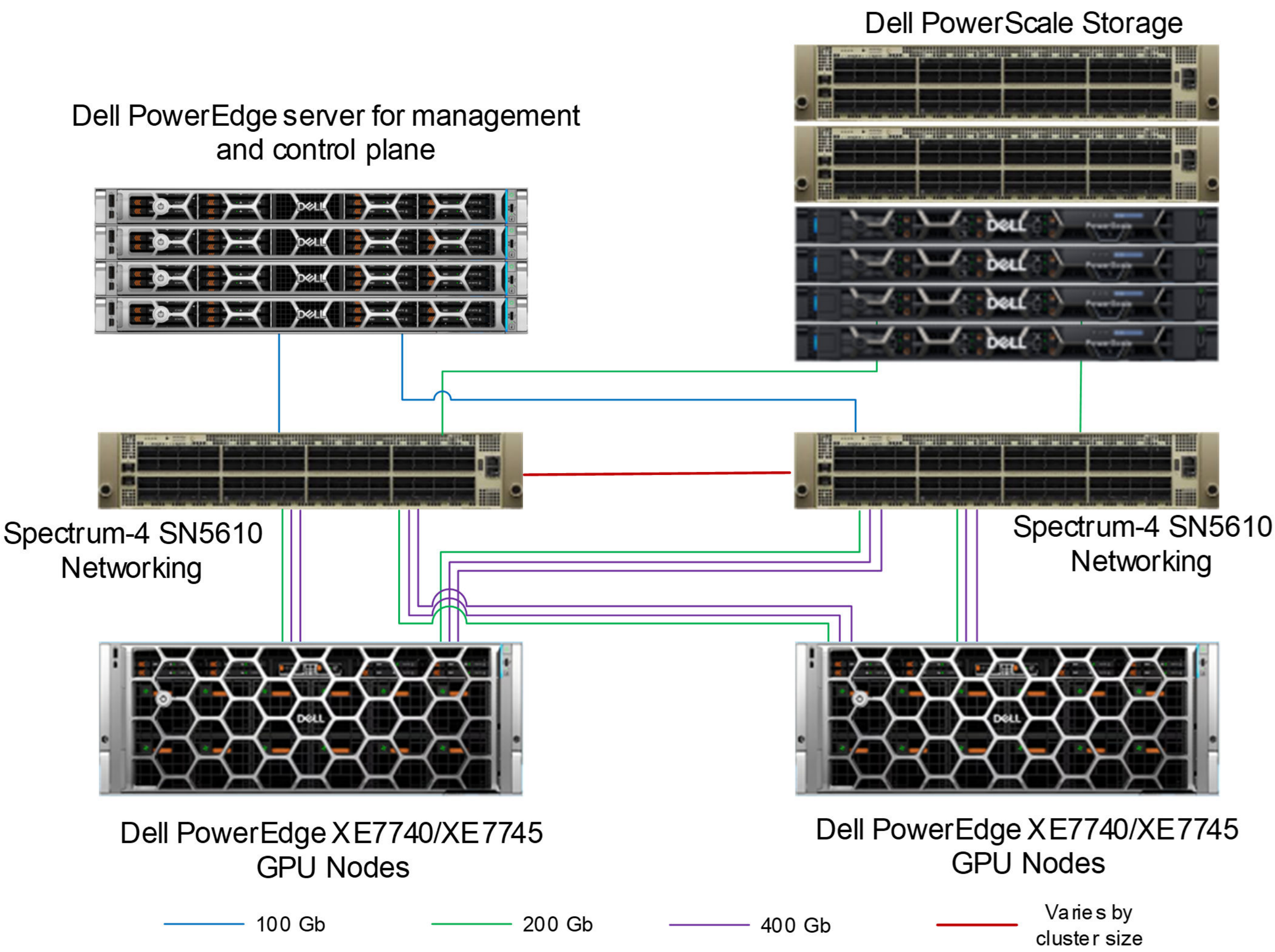

Figure 2. Network configuration for a PowerEdge XE7740/XE7745-based cluster

Using similar color coding, the following figure shows the network configuration of PowerEdge XE9680-based AI Platform.

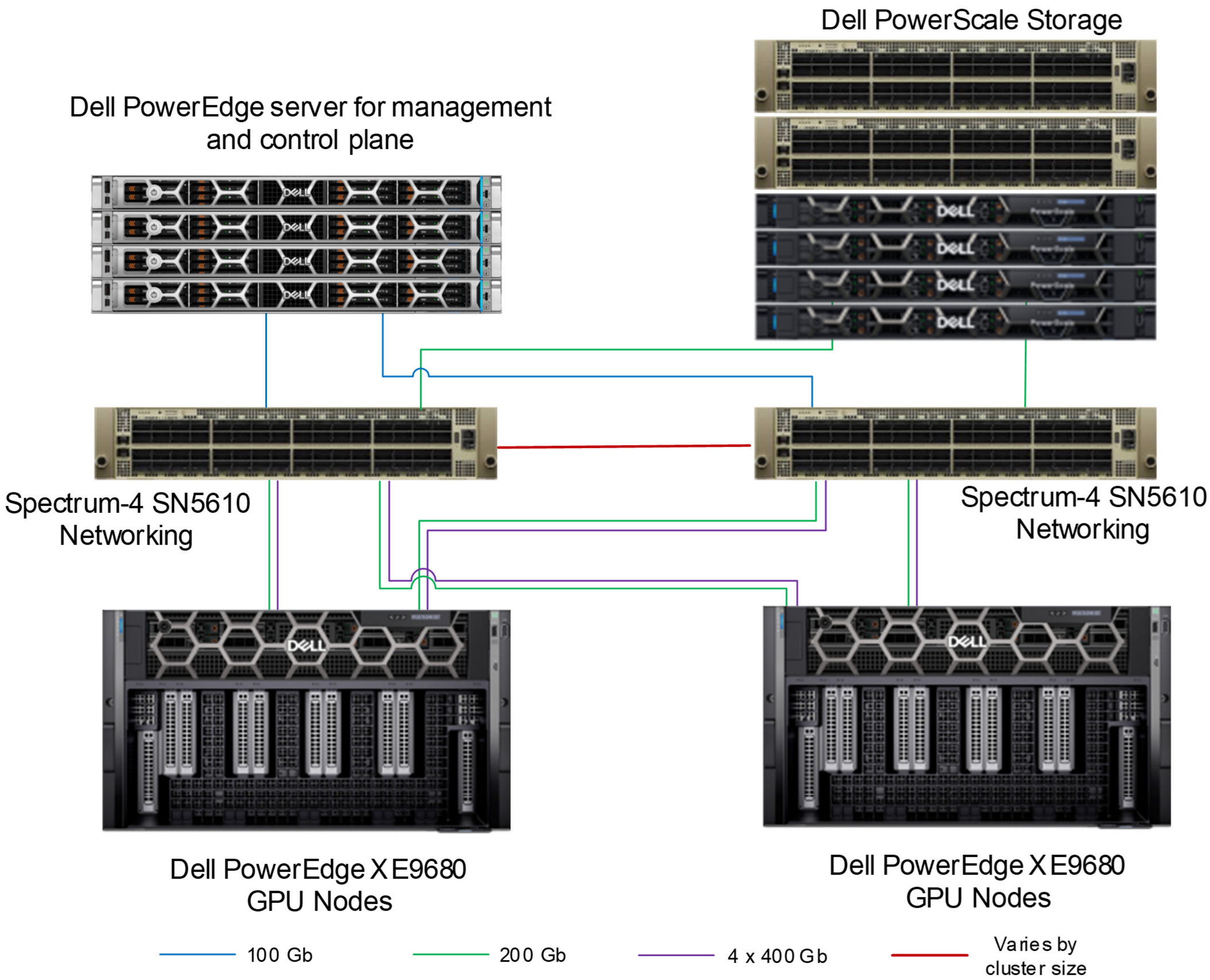

Figure 3. Network configuration for a PowerEdge XE9680-based cluster

Network architecture for PowerEdge Datacenter servers for inference only

Building on the GPU-based architecture described in the previous section, the network design for PowerEdge Datacenter Servers maintains the same high-performance principles while removing GPU Ethernet connectivity. This configuration uses PowerEdge R770, R7725, and R7715 servers as compute nodes, connected using 100 GbE links for storage and access.

using ConnectX-6 network adapters. The architecture continues to leverage dual SN5610 switches for EVPN-based network virtualization and redundancy, while PowerScale F710 storage remains integrated through 200 GbE connections. Out-of-band management is provided via the SN2201 GbE switch at 1 GbE, and 100 GbE inband links support control plane operations. This streamlined design delivers the required connectivity for inference only workloads, without the additional GPU networking layer.

While this architecture does not follow any NVIDIA Enterprise Reference Architecture (ERA) in its entirety, it adopts key principles from ERA for network configuration. ERA typically requires the use of NVIDIA BlueField-3 (BF3) adapters; however, these are not included in the reference design described here. Although PowerEdge servers support BF3, using these adapters consumes PCIe slots that could otherwise be allocated for GPUs. Therefore, for this configuration, we recommend using NVIDIA ConnectX-6 adapters to optimize slot utilization and maintain high network performance.

NOTE: A separate figure is not included because this architecture is nearly identical to the GPU-based configuration shown earlier, except for the removal of GPU Ethernet connectivity.

Logical network architecture

The logical network architecture for the Dell AI Factory is built on a collapsed, two-switch EVPN-VXLAN fabric that provides both the high-performance GPU backend and the virtualized enterprise network overlay for storage, management, and application traffic. Although the design uses only two Spectrum-X switches, the logical structure still follows modern data-center networking principles: EVPN acts as the control plane that distributes MAC and IP reachability information, enabling efficient forwarding for east-west and north-south traffic without relying on broadcast flooding. VXLAN provides scalable L2/L3 virtualization on top of an IP underlay, allowing the converged fabric to host multiple traffic domains—GPU, storage, management, and services—without requiring separate physical networks. Because the switches operate as the network virtualization edge (VTEP), they encapsulate and decapsulate VXLAN traffic, perform ARP suppression to reduce broadcast traffic, and support active-active multihoming, allowing compute and storage nodes to use both switches simultaneously for improved availability and bandwidth utilization. The IP underlay uses BGP for stable reachability, fast convergence, and clean separation between overlay control and physical transport. Together, these capabilities create a logical network architecture that is simple to manage, resilient to failures, and optimized for the performance characteristics of generative AI workloads

Cables and optics

NVIDIA's Ethernet Switch Configurator can be used to help identify and order the appropriate cables and optics for your AI fabrics. After entering your topology information into the form, it lists the available product options for your environment, along with part numbers and other information.

For the network design that is shown in Figure 8, customers can either use Direct Attached Copper (DAC) or fiber.

- For the backend network:
- **DAC option**—800 Gb/s Twin-port OSFP to 2x 400G QSFP cables are required. The 800 Gb OSFP connects to the Spectrum-X SN5610 switch while the 400G QSFP112 connects to BlueField-3 B3140H. See an example DAC Splitter cable product specification for more information.
- **Fiber option**—NVIDIA 800 Gb/s OSFP twin port transceiver can be used on the switch side, NVIDIA 400 Gbps QSFP112 single port transceiver on the BlueField-3 B3140H and NVIDIA passive fiber cable (MPO12 APC to MPO12) can be used to connect them.
- For the frontend network:
- **DAC option**—400 Gb/s QSFP-DD to QSFP112 cables are required. The 400G QSFP-DD connects to Spectrum-X SN5610, while the QSFP112 connects to BlueField-3 B3220. For more information, see the list of supported transceivers and cables.
- **Fiber option**—NVIDIA 400 Gbps QSFP-DD transceiver can be used on the switch side and 400 Gbps QSFP112 transceiver on the BlueField-3 B3220. NVIDIA passive fiber cable, MMF, MPO12 APC to 2 x MPO12 APC can be used to connect them.

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Solution validation

The Dell AI Platform for Generative AI simplifies and accelerates the deployment of complex infrastructure for generative AI by providing the reference architecture and configurations. It helps customers by reducing the guesswork and potential risks that are associated with designing and implementing initial custom solutions.

Topics:

- System configuration
- NCCL validation
- Inference validation
- Run:ai validation

System configuration

The following tables list the system configurations and software stack that are used for the validation efforts in this design.

Table 8. System configuration

Component	PowerEdge XE7740/ XE7745 configuration	PowerEdge XE9680 configuration	PowerEdge R770, R7725 and R7715 configuration
Compute server for model customization	4 x PowerEdge XE7740/ XE7745 servers	4 x PowerEdge XE9680 servers	1 x PowerEdge R770 1 x PowerEdge R7725 1 x PowerEdge R7715
GPUs per server	8 x NVIDIA H200 NVL GPUs or 8 x NVIDIA RTX Pro 6000	8 x NVIDIA H200 SXM GPUs	2 x NVIDIA RTX PRO 6000 or 2 x NVIDIA L40S GPUs (R770 and R7725) 2 x NVIDIA H200NVL (R7715)
Backend network adapters	4 x NVIDIA Bluefield-3 Single Port 400 GbE QSFP112 PCIe FH (B3140H)	8 x NVIDIA Bluefield-3 Single Port 400 GbE QSFP112 PCIe FH (B3140H)	None
Frontend network adapters	1 x NVIDIA Bluefield-3 Dual Port 200 GbE QSFP112 PCIe FH (B3220)	1 x NVIDIA Bluefield-3 Dual Port 200 GbE QSFP112 PCIe FH (B3220)	1 x NVIDIA ConnectX-6 Dx Dual Port 100GbE
Storage	PowerScale F710	PowerScale F710	PowerScale F710
Network switches	2 x NVIDIA Spectrum SN5610 switches 1 x NVIDIA Spectrum SN2201	2 x NVIDIA Spectrum SN5610 1 x NVIDIA Spectrum SN2201	2 x NVIDIA Spectrum SN5610 1 x NVIDIA Spectrum SN2201

Table 9. Software components and versions

Component	Details
Operating system	Ubuntu 24.04
AI software platform	NVIDIA AI Enterprise version 7.3
Cluster management	NVIDIA Base Command Manager Essentials 11.25.08
Kubernetes cluster	Version 1.34 with:

Table 9. Software components and versions (continued)

Component	Details
	Network Operator: 25.10.0 GPU Operator: 25.10.1 Dell CSM Operator: 1.16

NCCL validation

To validate the performance and operation of our GPU cluster deployed in a Kubernetes environment, we ran NVIDIA's nccl-tests within containerized pods across multiple nodes. These tests were orchestrated using MPI to benchmark collective communication operations such as AllReduce, Broadcast, and AllGather, allowing us to assess bandwidth, latency, and scalability under distributed workloads. Beyond verifying NCCL functionality, the tests also served to validate the underlying network stack, including backend networking configurations, SR-IOV setup, and RDMA enablement. This comprehensive validation ensured that both the software and hardware layers of our infrastructure were correctly tuned for high-performance GPU communication, laying a solid foundation for production-scale deep learning workloads.

Inference validation

As part of inference validation, the primary objective was to ensure the robustness, scalability, and performance of the infrastructure across a diverse set of generative AI workloads. This validation effort focused on deploying and benchmarking a range of large language and vision models—including Llama 3, Phi-3, GPT-OSS, and Gemma—across multiple GPU configurations and cluster topologies. The validation used NVIDIA AI Enterprise tools, including the NIM Operator and NIM containers, to streamline model deployment and inference. Where applicable, NVIDIA's LLM-specific NIM containers were prioritized for their optimized performance, with fallback to NIM Multi-LLM or vLLM-based stacks when required. Tests were conducted using both single-GPU and multi-GPU configurations, depending on the model's computational requirements. This structured and scalable approach ensures that the Dell AI Platform with NVIDIA can support a wide spectrum of inference scenarios, aligning with enterprise-grade performance expectations and enabling customers to confidently deploy production-ready AI solutions.

Run:ai validation

We deployed and validated Run:ai on Dell AI Platform with NVIDIA. We then validated the following Run:ai version 2.21.19 capabilities:

GPU slicing—We conducted a series of tests using fractional GPU allocations across three workload types: development workspace, inference, and training. We created a workspace using 10 percent of an NVIDIA H200 GPU with Jupyter Notebook as the development environment. For inference, we deployed the Llama 3.1 8B model using 35 percent of a GPU, authenticated by using Hugging Face, and confirmed model responsiveness with curl. For training, we allocated 50 percent of a GPU and used JupyterLab for interactive development. We monitored each workload through the Run:ai dashboard and verified GPU use with nvidia-smi, confirming accurate resource allocation and efficient multitenant GPU sharing.

Memory swap—We enabled the feature on Dell infrastructure by configuring CPU memory limits (250Gi), reserving 4GiB of GPU memory for nonswappable allocations, and labeling nodes to accept swap-enabled workloads. We created a dynamic compute resource with 30 percent minimum GPU memory allocation and sufficient CPU resources to support multiple concurrent models. Using this setup, we deployed three large language models—Llama 3.1 8B, Gemma 2 9B, and Phi-3.5 MoE—each of which typically require a dedicated NVIDIA H200 GPU. Through Run:ai's memory swap, we successfully ran all three models on a single GPU, dynamically swapping them between GPU and CPU memory. A web UI was used to interact with each model, demonstrating that while the swap introduces a few seconds of latency, it is a viable solution for low-demand environments.

Node pool—We created a dedicated node pool to manage a group of nodes with shared characteristics—specifically, nodes equipped with NVIDIA GPUs. Using the Run:ai platform, we defined the node pool using the Resources tab and associated it with a project by editing the project settings and assigning the node pool, which enabled scheduling of up to eight GPUs. We then validated the setup by deploying an NVIDIA NIM-based inference workload. This task involved configuring NGC credentials, selecting the appropriate node pool during workload creation, and confirming successful workload scheduling and execution.

This validation demonstrated the effectiveness of node pools in targeting specific hardware resources for optimized workload placement.

Scheduler—We tested the scheduler's ability to manage AI workloads with fairness, efficiency, and support for complex scheduling scenarios. We evaluated key features such as quota enforcement, overquota scheduling, preemption, and priority-based resource allocation. Our tests included deploying training and inference jobs under various quota conditions across multiple node pools. We confirmed that the scheduler correctly handled over-quota requests when idle resources were available, enforced nonpreemptive behavior for inference workloads, and prioritized high-importance jobs by preempting lower-priority ones. The scheduler’s queue fairness mechanism ensured equitable job distribution across projects, even when job submission volumes varied. These validations demonstrated the scheduler’s effectiveness in optimizing GPU utilization, maintaining workload fairness, and supporting high availability for critical inference services.

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Conclusion

Topics:

- Summary
- We value your feedback

Summary

The Dell AI Factory with NVIDIA provides a cohesive, validated, and enterprise-ready foundation for building and operationalizing generative AI at scale. By combining optimized compute platforms, a high-performance Spectrum-X Ethernet fabric, and a full-stack software ecosystem rooted in NVIDIA AI Enterprise, the solution delivers predictable performance and operational efficiency across training, fine-tuning, and inference workloads. The modular design allows organizations to adapt the architecture to their existing environments, integrate their preferred management and storage platforms, and evolve their AI capabilities as requirements grow.

Through alignment with NVIDIA's Enterprise Reference Architecture and the integration of robust orchestration, lifecycle management, and virtualization technologies, the Dell AI Factory offers a streamlined path from initial deployment to production-grade operation. This design guide provides the architectural direction, configuration guidance, and best practices needed to confidently build and scale a generative AI platform that remains manageable, efficient to operate, and capable of supporting evolving enterprise demands.

We value your feedback

Dell Technologies and the authors of this document welcome your feedback on this document and the information that it contains. Contact the Dell Technologies Solutions team by email.

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References

Topics:

- Dell Technologies documentation
- NVIDIA documentation

Dell Technologies documentation

The following Dell Technologies documentation provides additional and relevant information. Access to these documents depends on your login credentials. If you do not have access to a document, contact your Dell Technologies representative:

- Dell XE9680 Specification Sheet
- Dell R760xa Specification Sheet
- Dell PowerEdge Servers for AI
- The Dell AI Platform with NVIDIA
- Dell GenAI solutions
- Dell AI Services
- Dell Security and Trust Center
- Security Best Practices for Generative AI in the Enterprise
- Dell Info Hub

NVIDIA documentation

The following NVIDIA documentation provides additional and relevant information:

- NVIDIA AI Enterprise
- NVIDIA NeMo Framework
- NVIDIA Spectrum-X Networking Platform
- NVIDIA BlueField-3 Networking Platform
- NVIDIA Run:ai